

Alexander J. Hart

Aurora, Illinois

EDUCATION

Iowa State University; Ames, Iowa

Graduate Certificate, Power Systems Engineering, 2027

Northern Illinois University; DeKalb, Illinois

Master of Science, Electrical Engineering, 2023

Northern Illinois University; DeKalb, Illinois

Bachelor of Science, Electrical Engineering, 2022

WORK EXPERIENCE

Northrop Grumman; Rolling Meadows, Illinois

Systems Engineer

September 2023 – Present

- Design, develop and test real-time software and firmware algorithms for radar and GPS/INS navigation systems.
- Author system and software design documents detailing system behavior and functional requirements.
- Collaborate with system and software engineers to translate system and software requirements into C/C++ code.
- Simulate and evaluate a variety of real-time algorithms (e.g. Kalman filter) to define functional requirements.
- Develop tools in MATLAB and Python for engineering analysis and for automating engineering workflows.

Jet Propulsion Laboratory; Pasadena, California

Electrical Engineer Intern

June 2023 – August 2023

- Authored embedded C/C++ software for portable gas sensors to enable wireless data transmission over mesh networks.
- Developed a graphical user interface to process and display gas sensor and network health telemetry in real-time.

Electrical Engineer Intern

June 2022 – October 2022

- Conducted research in collaboration with NYSEARCH on UAV-based methane detection for natural gas infrastructure.
- Developed analytical tools in Python for processing simulated and real methane data.
- Participated in a controlled gas release in Upstate New York to collect real methane data for algorithm testing.

Northern Illinois University; DeKalb, Illinois

Graduate Research Assistant

November 2022 – May 2023

- Conducted research in collaboration with NYSEARCH on UAV-based methane detection for natural gas infrastructure.
- Evaluated the performance of a Bayesian estimation algorithm which estimated the location of a gas leak using a UAV.

Student Intern

June 2021 – August 2021

- Developed analytical tools in Python for processing simulated mesh network data.

PROJECTS

Power systems analysis (personal project)

June 2026 - Present

- Study a mix of power systems analysis topics including line, generator, and transformer modeling and power flows. Develop and run related simulations in PSS/E.

Simulation of a DFIG-based Wind Turbine (ISU EE 5590)

May 2026

- Modeling and simulation of a DFIG-based wind turbine and power electronics for grid integration. Written in MATLAB.

Simulation of EMS algorithms (ISU EE 5530)

December 2025

- Developed a mix of EMS algorithms including power flow, optimal power flow, economic dispatch, contingency analysis, and security-constrained optimal power flow. Written in MATLAB using MATPOWER.

ACTIVITIES

Abbott Power Plant / Solar Farm 2.0

July 2026

- Self-funded visit to the University of Illinois Abbott Power Plant and Solar Farm 2.0 to learn about how these unique generation facilities provide power to 60,000 students and faculty on campus.

Power and Energy Conference at Illinois (PECI)

April 2026

- Self-funded visit to PEGI 2026 to learn about emerging research in power and energy.

CERTIFICATIONS

Fundamentals of Engineering, Electrical and Computer Exam

June 2026 - Present

- Exam scheduled August 31, 2026